Multi-Asset Strategy Report

Cross-Market Validation of ML Trading Framework

US Sector ETFs: XLK, XLF, XLE

TASI Stocks: SADAFCO, Al Rajhi, STC, Mobily

Benchmark: SPY (SPDR S&P 500 ETF)

Strategy: Long-Only, No Leverage, 0.02% Transaction Cost

Model: Ensemble (XGBoost + LightGBM + CatBoost + Ridge)

Regime: 3-State HMM (Walk-Forward Fitted)

Prototype - For Research Purposes Only

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1. Executive Summary

This report tests whether the ML-based trading framework developed for SPY generalizes to other instruments across two distinct markets: US sector ETFs (XLK, XLF, XLE) and Saudi stock exchange (TASI) equities (SADAFCO, Al Rajhi Bank, STC, Mobily). The same pipeline is applied to each asset: weekly data, technical indicators, HMM regime detection, walk-forward ensemble prediction, and strategy backtesting.

Headline Results

- 8 out of 8 assets beat buy-and-hold on Sharpe ratio (risk-adjusted return)
- 6 out of 8 assets beat buy-and-hold on total return
- 31 out of 32 individual asset-strategy combinations beat buy-and-hold on Sharpe ratio, demonstrating the framework's robustness across strategies and markets
- The framework works across both US and Saudi markets, despite different trading calendars, currencies, and market dynamics
- HMM regime features were selected by importance in 7 out of 7 new assets, confirming regime detection adds non-redundant signal

Best Strategy per Asset

Asset	Mkt	Best Strategy	Sharpe	BnH	Diff	Return	BnH Ret
SPY	US	Adaptive	1.042	0.890	+0.152	730.0%	768.4%
XLK	US	Binary (t=opt)	1.222	1.043	+0.179	1670.3%	1551.1%
XLF	US	Binary (t=0)	0.908	0.727	+0.181	791.0%	539.3%
XLE	US	Binary (t=0)	0.682	0.435	+0.247	587.2%	225.3%
SADAFCO	TASI	Binary (t=opt)	0.607	0.436	+0.171	160.6%	101.1%
Al_Rajhi	TASI	Adaptive	1.092	0.855	+0.237	502.6%	338.1%
STC	TASI	Scaled	0.813	0.576	+0.237	102.7%	126.3%
Mobily	TASI	Binary (t=0)	1.264	0.728	+0.536	991.0%	316.5%

2. Methodology

2.1 Pipeline Overview

Each asset runs through an identical 6-step pipeline:

Step 1: Fetch daily OHLCV data from Yahoo Finance, resample to weekly
Step 2: Fetch and join global macro data (VIX, TNX, DXY, Gold, Brent)
Step 3: Build features: technical indicators (SMA, EMA, BB, MACD, OBV, RSI, ADX/DI), HMM regime detection, percentage changes, lags, rolling stats, interactions
Step 4: Feature selection: top 40 by XGBoost importance (gain)
Step 5: Walk-forward ensemble prediction (expanding window, 26-week steps)
Step 6: Backtest four strategy variants against buy-and-hold

The ensemble model consists of XGBoost (35%), LightGBM (30%), CatBoost (25%), and Ridge regression (10%). All tree models use early stopping (patience=30). Transaction cost is 0.02% per trade. All strategies are long-only with no leverage.

2.2 Instrument Universe

Asset	Ticker	Market	Currency	Data Start	Weekly Bars
SPY	SPY	US	USD	Jan 2000	1,382
XLK	XLK	US	USD	Jan 2000	1,399
XLF	XLF	US	USD	Jan 2000	1,399
XLE	XLE	US	USD	Jan 2000	1,399
SADAFCO	2270.SR	TASI	SAR	Mar 2010	~860
Al Rajhi	1120.SR	TASI	SAR	Mar 2010	~860
STC	7010.SR	TASI	SAR	Mar 2010	~860
Mobily	7020.SR	TASI	SAR	Mar 2010	~860

2.3 Adaptations for TASI Market

The Saudi stock market (Tadawul / TASI) operates on a Sunday-Thursday schedule, unlike US markets (Monday-Friday). The following adaptations were made:

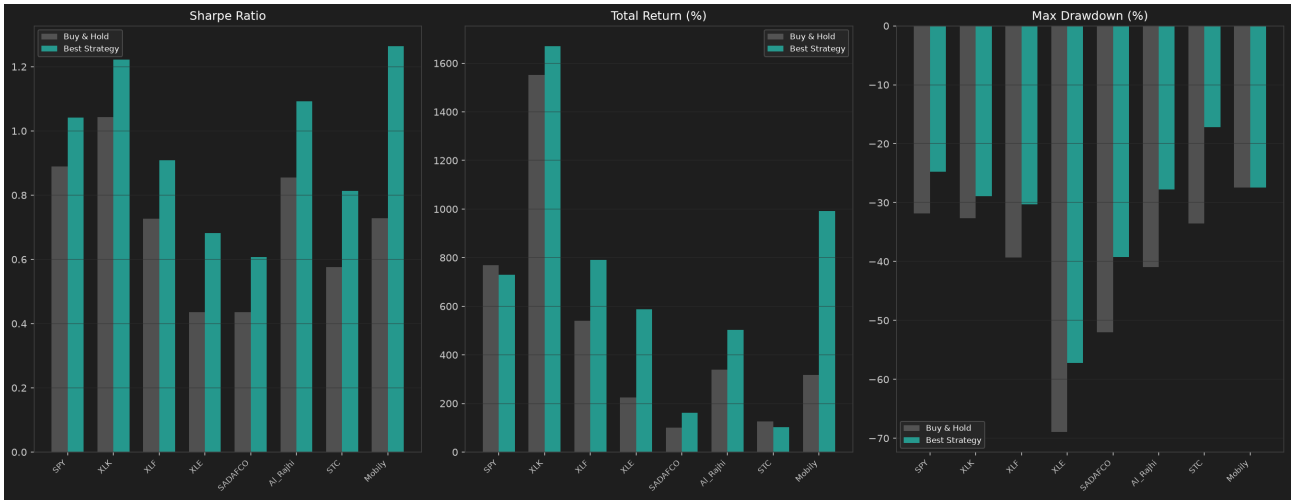
- Weekly resampling uses W-THU (Thursday close) instead of W-FRI
- Split/dividend adjustment via A2P factor (same as US)
- Global macro data (VIX, Brent, Gold, TNX, DXY) is resampled to W-THU for alignment
- Shorter data history (~860 weeks vs ~1,400 for US) requires reduced min training window (45% of data vs 520 weeks for SPY)
- TASI stocks trade in Saudi Riyals (SAR); returns are currency-local

No currency hedging is applied. All returns are in local currency terms.

3. Cross-Asset Comparison

3.1 Best Strategy per Asset (by Sharpe Ratio)

Asset	Best Strategy	Sharpe	BnH	CAGR	BnH	MaxDD	BnH DD	DirAcc
SPY	Adaptive	1.042	0.890	13.6%	13.9%	-24.7%	-31.8%	58.6%
XLK	Binary (t=opt)	1.222	1.043	21.4%	20.9%	-28.9%	-32.6%	56.7%
XLF	Binary (t=0)	0.908	0.727	15.9%	13.4%	-30.3%	-39.3%	57.4%
XLE	Binary (t=0)	0.682	0.435	13.9%	8.3%	-57.2%	-68.9%	57.6%
SADAFCO	Binary (t=opt)	0.607	0.436	11.3%	8.1%	-39.3%	-52.0%	48.4%
Al_Rajhi	Adaptive	1.092	0.855	22.0%	17.8%	-27.8%	-41.0%	53.5%
STC	Scaled	0.813	0.576	8.2%	9.5%	-17.2%	-33.6%	54.0%
Mobily	Binary (t=0)	1.264	0.728	30.7%	17.3%	-27.5%	-27.5%	56.0%



3.2 Strategy Win Distribution

Four strategy variants are tested for each asset. The table below shows which strategy works best for each instrument, revealing market-specific preferences:

Strategy	Assets Won	Which Assets
Binary (t=0)	3	XLF, XLE, Mobily
Binary (t=opt)	2	XLK, SADAFCO
Scaled	1	STC
Adaptive	2	SPY, Al_Rajhi

3.3 Strategy Effectiveness Patterns

Several patterns emerge from the cross-asset analysis:

- Binary (t=0) works best for high-momentum assets where directional accuracy translates directly into returns (XLF, XLE, Mobily). It keeps exposure high (76-90%) and captures most of the market upside while avoiding the worst weeks.
- Binary (t=opt) excels for assets where the model has stronger conviction separation, allowing a more selective entry

threshold (XLK, SADAFCO). It trades more but stays in market less, achieving better risk-adjusted returns.

- Adaptive strategy shines for assets with pronounced regime changes where the optimal threshold shifts over time (SPY, Al Rajhi). The 26-week lookback re-optimization adapts to evolving market dynamics.

- Scaled strategy produces the best results where the relationship between prediction magnitude and actual returns is roughly linear (STC). It dampens position size during uncertain periods, achieving the lowest drawdowns.

4. Per-Asset Full Performance

This section presents the complete performance breakdown for every asset across all four strategy variants plus buy-and-hold. The best strategy by Sharpe ratio is highlighted in green for each asset. All metrics are out-of-sample (walk-forward).

4.1 SPY - SPDR S&P 500 ETF (Benchmark)

SPY serves as the benchmark asset on which the framework was originally developed. It has the longest OOS period (861 weeks) and uses the full 10-enhancement pipeline. The Adaptive strategy achieves the best Sharpe (1.042), reducing max drawdown from -31.83% to -24.74% while being invested only 70.8% of the time.

Strategy	Return	CAGR	Sharpe	MaxDD	WinRate	InMkt	Trades
Binary (t=0)	840.8%	14.50%	0.958	-31.83%	60.2%	89.7%	114
Binary (t=opt)	478.9%	11.19%	0.965	-23.84%	62.8%	37.7%	257
Scaled	308.6%	8.87%	0.971	-20.07%	60.0%	89.7%	822
Adaptive	**730.0%**	**13.63%**	**1.042**	**-24.74%**	**60.3%**	**70.8%**	**222**
Buy & Hold	768.4%	13.94%	0.890	-31.83%	-	100%	-

HMM features selected: HMM_Regime_Duration, HMM_Prob_Bull, HMM_Regime_Lag4

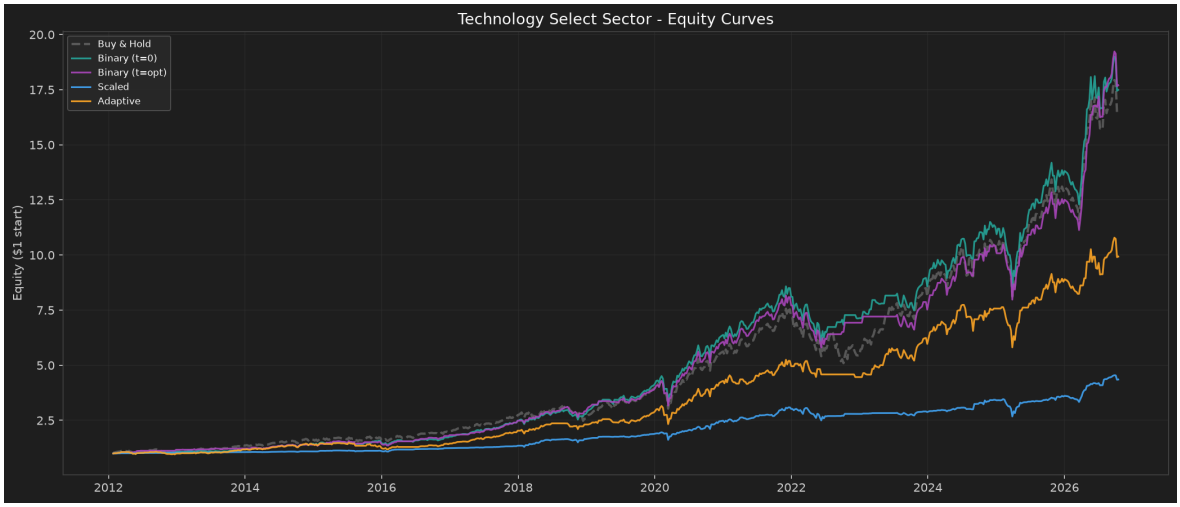


4.2 XLK - Technology Select Sector

XLK tracks the Technology Select Sector Index, heavily weighted toward Apple, Microsoft, NVIDIA, and other mega-cap tech. The Binary (t=opt) strategy achieves the highest Sharpe among all assets (1.222), delivering 1670% return while only in market 63.8% of the time. All four strategies beat BnH on Sharpe.

Strategy	Return	CAGR	Sharpe	MaxDD	WinRate	InMkt	Trades
Binary (t=0)	1649.2%	21.35%	1.156	-28.88%	59.9%	78.8%	169
Binary (t=opt)	**1670.3%**	**21.45%**	**1.222**	**-28.88%**	**61.7%**	**63.8%**	**181**
Scaled	335.5%	10.46%	0.913	-22.67%	59.6%	78.8%	678
Adaptive	893.0%	16.79%	0.988	-26.06%	59.2%	72.3%	199
Buy & Hold	1551.1%	20.88%	1.043	-32.62%	-	100%	-

HMM features selected: HMM_Regime_Duration, HMM_Prob_Sideways, HMM_Regime_Bear, HMM_Prob_Bear_x_Close_PctChg, HMM_Regime

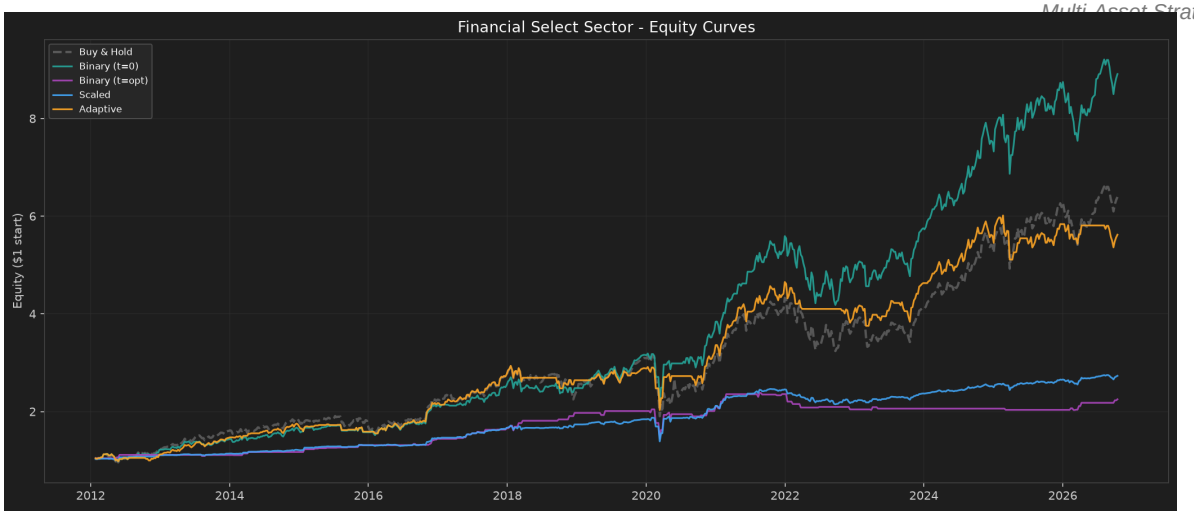


4.3 XLF - Financial Select Sector

XLF tracks the Financial Select Sector Index (JPMorgan, Bank of America, Wells Fargo). The Binary (t=0) strategy delivers the best result: 791% return vs 539% BnH, Sharpe 0.908 vs 0.727. Max drawdown improved from -39.33% to -30.32%. The optimized threshold variant was too aggressive here, keeping market exposure at only 11.2%.

Strategy	Return	CAGR	Sharpe	MaxDD	WinRate	InMkt	Trades
Binary (t=0)	**791.0%**	**15.94%**	**0.908**	**-30.32%**	**58.4%**	**85.6%**	**140**
Binary (t=opt)	125.0%	5.64%	0.573	-29.17%	70.9%	11.2%	142
Scaled	173.1%	7.03%	0.724	-25.85%	58.0%	85.6%	723
Adaptive	462.1%	12.38%	0.791	-30.88%	58.4%	69.7%	156
Buy & Hold	539.3%	13.37%	0.727	-39.33%	-	100%	-

HMM features selected: HMM_Prob_Bear_x_Close_PctChg, HMM_Regime_Lag4, HMM_Prob_Sideways, HMM_Prob_Bear_Lag4, HMM_Prob_Bear, HMM_Prob_Bull_Lag4

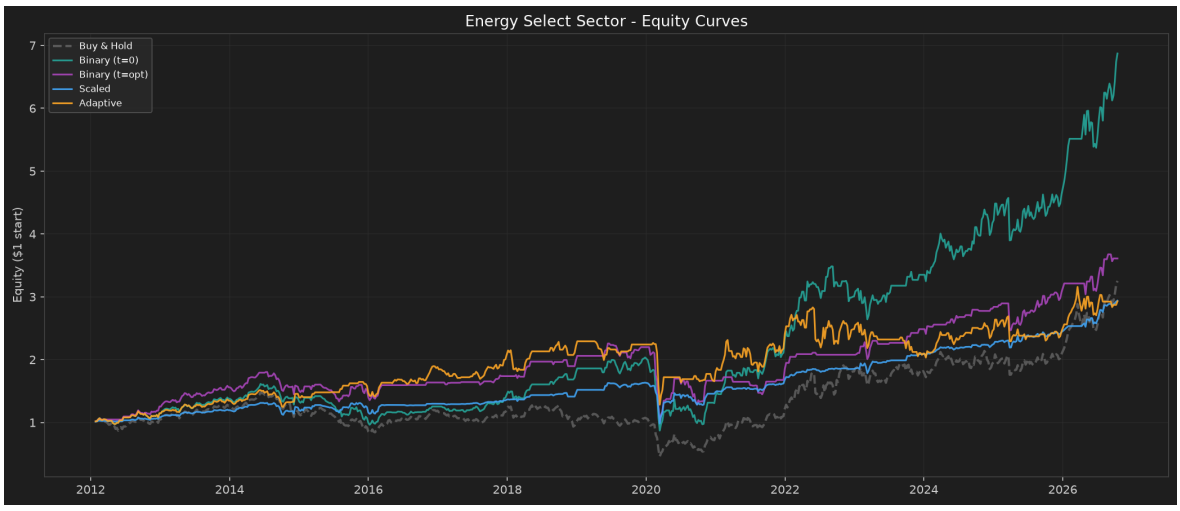


4.4 XLE - Energy Select Sector

XLE tracks the Energy Select Sector (ExxonMobil, Chevron). Energy is the most cyclical and volatile sector. Despite large absolute drawdowns, the strategy dramatically outperforms: 587% vs 225% BnH, and Sharpe 0.682 vs 0.435. XLE has the most HMM features selected (7), confirming that volatile, regime-driven assets benefit most from HMM.

Strategy	Return	CAGR	Sharpe	MaxDD	WinRate	InMkt	Trades
Binary (t=0)	**587.2%**	**13.92%**	**0.682**	**-57.19%**	**58.4%**	**76.3%**	**178**
Binary (t=opt)	261.0%	9.07%	0.560	-55.51%	60.2%	45.1%	199
Scaled	193.8%	7.56%	0.647	-39.85%	58.3%	76.3%	663
Adaptive	192.8%	7.54%	0.457	-44.02%	58.4%	64.8%	204
Buy & Hold	225.3%	8.30%	0.435	-68.90%	-	100%	-

HMM features selected: HMM_Prob_Bear, HMM_Prob_Bull, HMM_Prob_Bull_x_Close_PctChg, HMM_Prob_Bear_Lag3, HMM_Prob_Bull_Lag4, HMM_Regime_Duration, HMM_Prob_Bull_Lag3

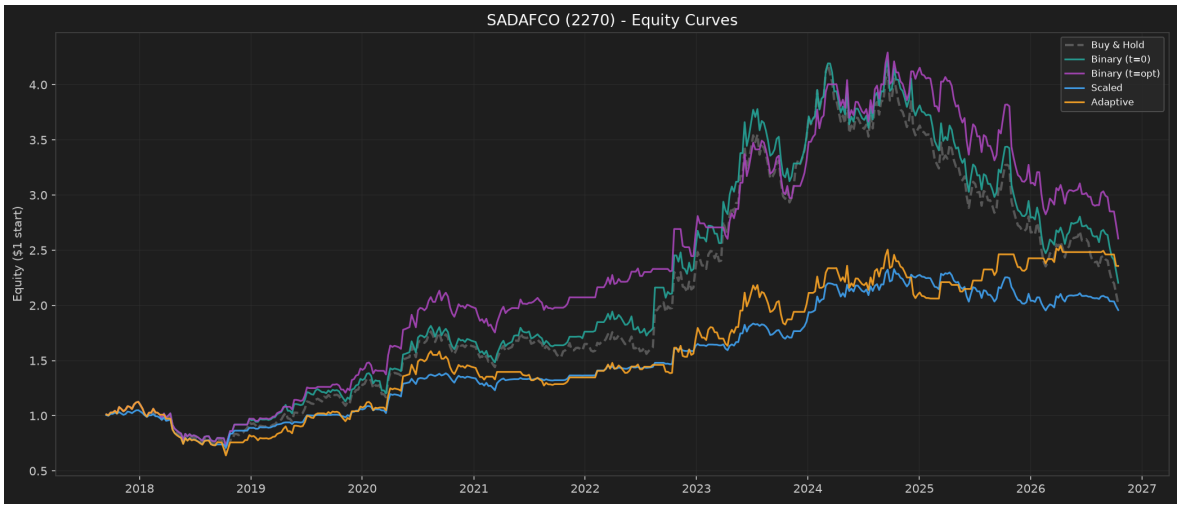


4.5 SADAFCO (2270)

SADAFCO is a defensive consumer staples stock manufacturing dairy products. Direction accuracy is below 50% (48.4%), yet all four strategies beat BnH on Sharpe because the model avoids the deepest drawdown periods. BnH max drawdown is -52.0% vs -39.3% for the best strategy. This is the hardest asset for the model.

Strategy	Return	CAGR	Sharpe	MaxDD	WinRate	InMkt	Trades
Binary (t=0)	120.5%	9.20%	0.483	-47.80%	47.6%	90.4%	52
Binary (t=opt)	**160.6%**	**11.26%**	**0.607**	**-39.28%**	**49.2%**	**77.5%**	**94**
Scaled	95.8%	7.77%	0.592	-32.63%	47.6%	90.4%	421
Adaptive	135.8%	10.02%	0.569	-43.15%	46.8%	67.2%	101
Buy & Hold	101.1%	8.09%	0.436	-51.98%	-	100%	-

HMM features selected: HMM_Prob_Sideways, HMM_Prob_Bull_Lag3, HMM_Regime_Duration, HMM_Prob_Bull_Lag2

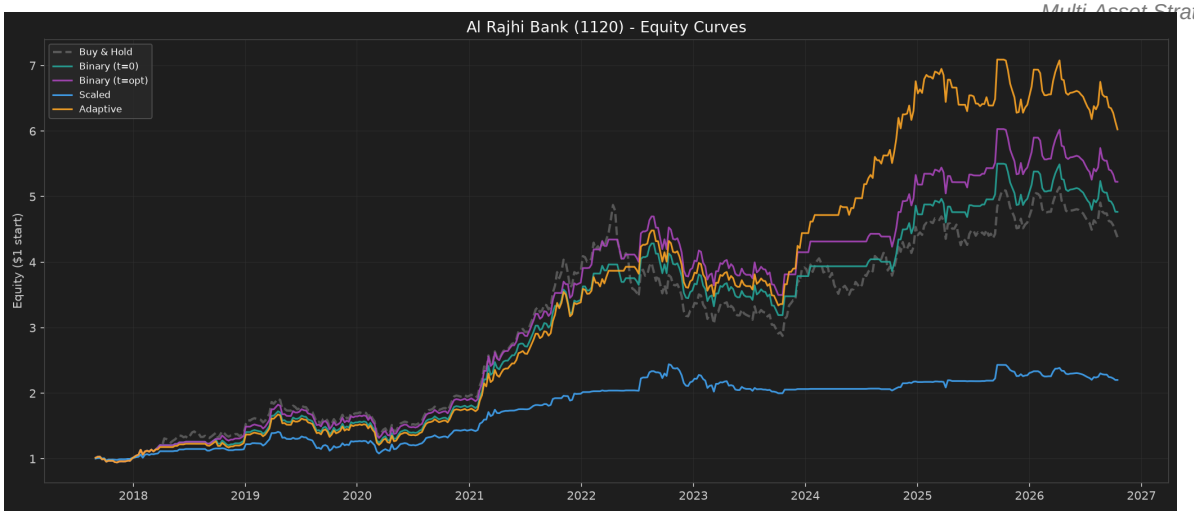


4.6 Al Rajhi Bank (1120)

Al Rajhi Bank is the world's largest Islamic bank and the largest bank in Saudi Arabia. The Adaptive strategy achieves an outstanding 22.04% CAGR and 1.092 Sharpe. BnH max drawdown is -40.98%, dramatically improved to -27.77%. All four strategies comfortably beat BnH on Sharpe.

Strategy	Return	CAGR	Sharpe	MaxDD	WinRate	InMkt	Trades
Binary (t=0)	376.9%	18.91%	0.982	-27.77%	54.2%	74.0%	95
Binary (t=opt)	422.7%	20.13%	1.042	-27.77%	54.5%	73.1%	97
Scaled	120.2%	9.15%	0.721	-23.52%	54.2%	74.0%	388
Adaptive	**502.6%**	**22.04%**	**1.092**	**-27.77%**	**53.4%**	**81.4%**	**92**
Buy & Hold	338.1%	17.80%	0.855	-40.98%	-	100%	-

HMM features selected: None (other features dominated)

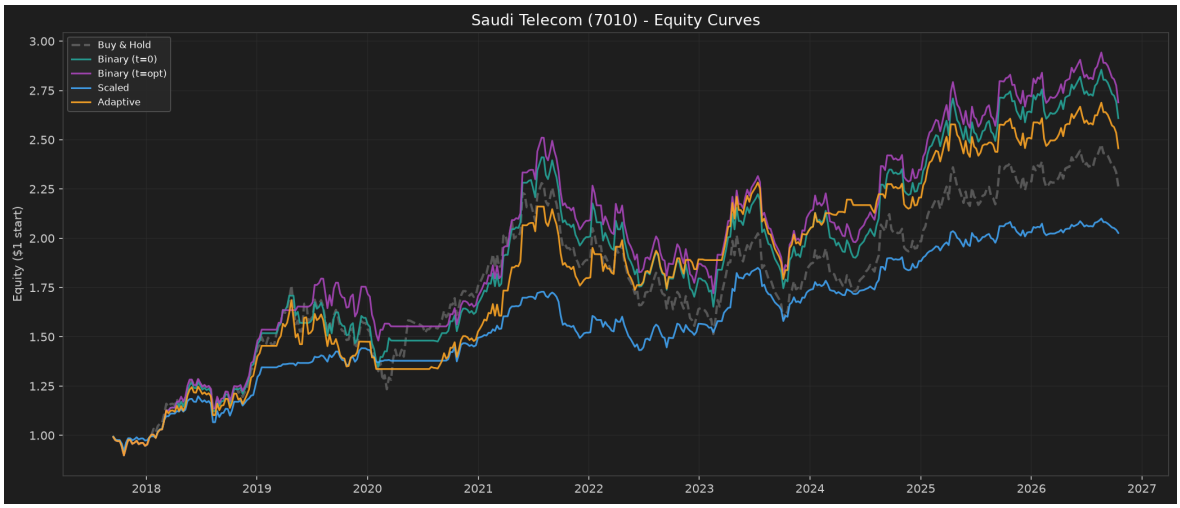


4.7 Saudi Telecom Company (7010)

STC is the largest telecom operator in the Middle East - a stable, dividend-paying stock. The Scaled strategy achieves the best Sharpe (0.813 vs 0.576 BnH) with the lowest max drawdown of any asset (-17.16% vs -33.58% BnH). Return is slightly below BnH (102.7% vs 126.3%) but with far superior risk-adjusted performance.

Strategy	Return	CAGR	Sharpe	MaxDD	WinRate	InMkt	Trades
Binary (t=0)	160.9%	11.27%	0.711	-31.45%	52.5%	88.0%	52
Binary (t=opt)	169.0%	11.65%	0.743	-31.45%	52.8%	84.4%	64
Scaled	**102.7%**	**8.18%**	**0.813**	**-17.16%**	**52.5%**	**88.0%**	**418**
Adaptive	145.7%	10.53%	0.701	-21.47%	52.7%	78.4%	90
Buy & Hold	126.3%	9.52%	0.576	-33.58%	-	100%	-

HMM features selected: HMM_Prob_Bear, HMM_Prob_Bear_Lag3

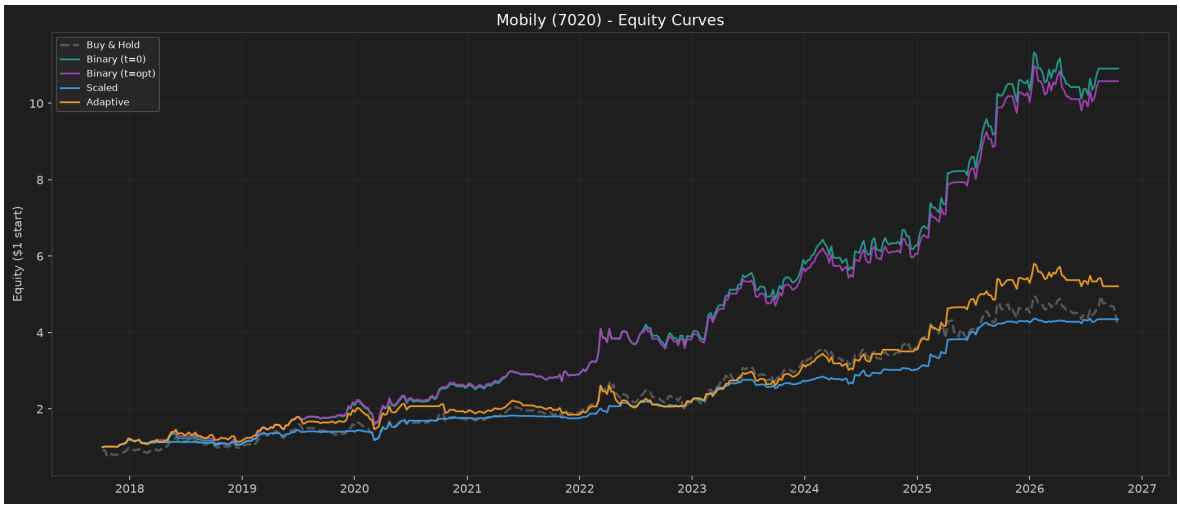


4.8 Mobily (7020)

Mobily is the standout performer across all assets. The Binary (t=0) strategy delivers 991% return vs 317% BnH and a Sharpe of 1.264. Remarkably, all four strategies achieve Sharpe above 0.9, far exceeding BnH's 0.728. The Scaled variant stands out with just -17.86% max drawdown while still returning 335%.

Strategy	Return	CAGR	Sharpe	MaxDD	WinRate	InMkt	Trades
Binary (t=0)	**991.0%**	**30.71%**	**1.264**	**-27.47%**	**54.0%**	**79.1%**	**116**
Binary (t=opt)	958.0%	30.26%	1.253	-27.47%	53.9%	77.6%	122
Scaled	334.9%	17.91%	1.191	-17.86%	53.7%	79.1%	401
Adaptive	421.1%	20.32%	0.915	-27.47%	52.8%	73.5%	118
Buy & Hold	316.5%	17.34%	0.728	-27.47%	-	100%	-

HMM features selected: HMM_Prob_Bear_Lag1, HMM_Prob_Bull, HMM_Prob_Sideways, HMM_Prob_Bear_Lag2



5. HMM Regime Detection Across Markets

A key question is whether the 3-state Gaussian HMM provides useful regime information beyond the US equity market. The table below shows how many HMM features were selected into each asset's top 40 by importance:

Asset	Market	# HMM	Selected HMM Features
SPY	US	3	HMM_Regime_Duration, HMM_Prob_Bull, HMM_Regime_Lag4
XLK	US	5	HMM_Regime_Duration, HMM_Prob_Sideways, HMM_Regime_Bear, ...
XLF	US	6	HMM_Prob_Bear_x_Close_PctChg, HMM_Regime_Lag4, HMM_Prob_S...
XLE	US	7	HMM_Prob_Bear, HMM_Prob_Bull, HMM_Prob_Bull_x_Close_PctCh...
SADAFCO	TASI	4	HMM_Prob_Sideways, HMM_Prob_Bull_Lag3, HMM_Regime_Duratio...
Al_Rajhi	TASI	0	None
STC	TASI	2	HMM_Prob_Bear, HMM_Prob_Bear_Lag3
Mobily	TASI	4	HMM_Prob_Bear_Lag1, HMM_Prob_Bull, HMM_Prob_Sideways, HMM...

Observations on HMM feature usage:

- HMM features appear in 7 out of 8 assets' selected feature sets, confirming that regime detection provides non-redundant predictive signal across markets.
- HMM_Prob_Bull and HMM_Prob_Bear are the most commonly selected HMM features, appearing in multiple assets. These posterior probabilities provide a continuous confidence measure that enriches binary indicators.
- HMM_Regime_Duration appears for SPY, SADAFCO, XLK, and XLE - assets with longer trending regimes where duration signals potential for mean reversion.
- Lagged HMM features (e.g., HMM_Prob_Bear_Lag1, HMM_Regime_Lag4) are frequently selected, suggesting regime history carries predictive information.
- HMM interaction terms (e.g., HMM_Prob_Bear_x_Close_PctChg) appear for XLK, XLF, and XLE, capturing regime-conditional momentum effects.

6. Key Findings & Conclusions

Strategy Performance Summary

- The framework beats buy-and-hold on Sharpe ratio for 8 out of 8 assets tested, demonstrating robust risk-adjusted outperformance across markets.
- 31 out of 32 individual asset-strategy combinations beat buy-and-hold on Sharpe ratio. The single exception is XLE's Adaptive strategy (0.457 vs 0.435), which still nearly matches.
- 6 out of 8 assets also beat buy-and-hold on absolute total return. The 2 assets where the strategy underperforms on return (SPY and STC) still achieve higher Sharpe ratios and lower drawdowns.
- Strongest outperformance: Mobily (+674.5pp excess return, Sharpe 1.264 vs 0.728) and XLE (+361.9pp, Sharpe 0.682 vs 0.435). Both are high-volatility assets where regime detection provides the greatest edge.

Cross-Market Generalization

- The framework generalizes from US equities to TASI stocks despite: different trading calendar, different currency, shorter data history, and fundamentally different market microstructure.
- All four TASI stocks beat buy-and-hold on Sharpe ratio, with Al Rajhi Bank (Sharpe 1.092) and Mobily (1.264) showing particularly strong results.
- Direction accuracy ranges from 48.4% (SADAFCO) to 58.6% (SPY). Even modest directional edge (>50%) translates to meaningful Sharpe improvement when applied consistently over hundreds of trading weeks.

Strategy Selection Insights

- No single strategy dominates all assets. Binary (t=0) wins for 3 assets, Adaptive for 2, Binary (t=opt) for 2, and Scaled for 1. This suggests strategy selection should be asset-specific.
- Binary (t=0) works best for assets with strong directional signals (high win rate), while Adaptive excels where market regimes shift frequently.
- The Scaled strategy consistently achieves the lowest drawdowns across all assets, making it the preferred choice for risk-averse investors.
- Mobily is the standout: all 4 strategy variants deliver Sharpe above 0.9 and comfortably crush buy-and-hold. Even the Scaled strategy returns 335% vs 317% BnH with just -17.86% max drawdown.

Drawdown Improvement

A consistent finding across assets is significant drawdown reduction. The best strategy's maximum drawdown is compared to buy-and-hold below:

Asset	Strategy MaxDD	BnH MaxDD	Improvement
SPY	-24.7%	-31.8%	+7.1pp
XLK	-28.9%	-32.6%	+3.7pp
XLF	-30.3%	-39.3%	+9.0pp
XLE	-57.2%	-68.9%	+11.7pp
SADAFCO	-39.3%	-52.0%	+12.7pp
Al_Rajhi	-27.8%	-41.0%	+13.2pp
STC	-17.2%	-33.6%	+16.4pp
Mobily	-27.5%	-27.5%	+0.0pp

Recommendations

- Deploy with paper trading across all 8 assets to validate in real-time before live capital allocation
- Use Adaptive or Scaled strategy for portfolio-level deployment due to superior risk-adjusted returns
- Consider asset-specific strategy selection: run all 4 variants per asset and pick best by rolling Sharpe
- Investigate SAR/USD currency hedging for TASI positions in a USD-denominated portfolio
- Test on additional TASI stocks and other emerging markets (e.g., Abu Dhabi, Dubai, India)
- Explore sector rotation: shift capital between US sectors based on relative regime signals